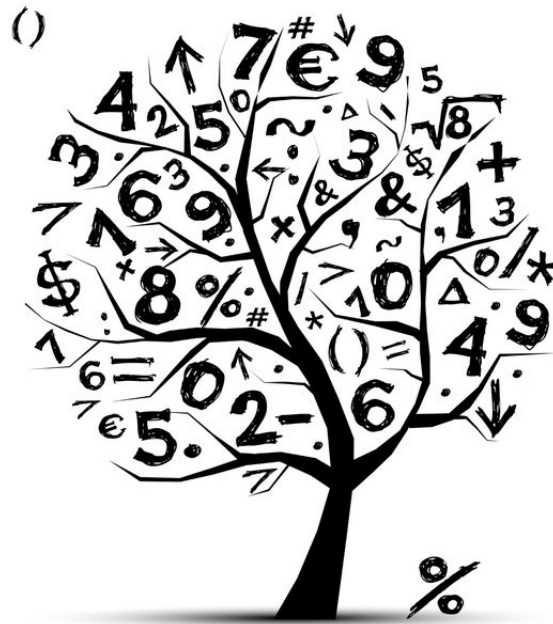


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Exercises: Inversion of matrices



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Exercise 1

Using the formula for the inverse of a 2×2 matrix, compute (if it exists) the inverse of

$$\mathbf{A} = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 4 & 2 \\ 3 & 4 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 2 & 4 \\ 3 & 6 \end{pmatrix}.$$

Recall that, if $ad - bc \neq 0$, $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

- $\det(\mathbf{A}) = 3 \cdot 2 - 5 \cdot 1 = 1$, hence

$$\mathbf{A}^{-1} = \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix}. \quad \text{Check: } \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 6-5 & -15+15 \\ 2-2 & -5+6 \end{pmatrix} = \mathbf{I}.$$

- $\det(\mathbf{B}) = 4 \cdot 4 - 2 \cdot 3 = 10$, hence

$$\mathbf{B}^{-1} = \frac{1}{10} \begin{pmatrix} 4 & -2 \\ -3 & 4 \end{pmatrix} = \begin{pmatrix} \frac{2}{5} & -\frac{1}{5} \\ -\frac{3}{10} & \frac{2}{5} \end{pmatrix}.$$

- $\det(\mathbf{C}) = 2 \cdot 6 - 4 \cdot 3 = 0$: the matrix $\mathbf{C}$ is singular and $\mathbf{C}^{-1}$ does not exist (the second row is $\frac{3}{2}$ times the first one).

Exercise 2

Compute the inverse of

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

with the method of Gauss–Jordan, and verify the result.

We apply elementary row operations to $(A | I)$:

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right)$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right) \quad (R_2 \leftarrow R_2 - R_1)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 2 & -1 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right) \quad (R_1 \leftarrow R_1 - R_2)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 2 & -1 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right) \quad (R_3 \leftarrow R_3 - R_2)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -2 & 1 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right) \quad (R_1 \leftarrow R_1 + R_3)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -2 & 1 \\ 0 & 1 & 0 & -2 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right) \quad (R_2 \leftarrow R_2 - R_3)$$

Hence

$$A^{-1} = \begin{pmatrix} 3 & -2 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix}.$$

Check (row by column): $AA^{-1} = \begin{pmatrix} 3-2 & -2+2 & 1-1 \\ 3-4+1 & -2+4-1 & 1-2+1 \\ -2+2 & 2-2 & -1+2 \end{pmatrix} = I.$

Exercise 3

Compute the inverse of

$$A = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix}$$

with the method of Gauss–Jordan.

Since $a_{11} = 0$, we start with a row swap. Later, we swap rows 2 and 3 to get a pivot equal to 1 (and avoid fractions):

$$\begin{aligned}
 & \left(\begin{array}{ccc|ccc} 0 & 2 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 2 & 3 & 1 & 0 & 0 & 1 \end{array} \right) \\
 & \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 2 & 1 & 1 & 0 & 0 \\ 2 & 3 & 1 & 0 & 0 & 1 \end{array} \right) \quad (R_1 \leftrightarrow R_2) \\
 & \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & -2 & 1 \end{array} \right) \quad (R_3 \leftarrow R_3 - 2R_1) \\
 & \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & -2 & 1 \\ 0 & 2 & 1 & 1 & 0 & 0 \end{array} \right) \quad (R_2 \leftrightarrow R_3) \\
 & \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 0 & 3 & -1 \\ 0 & 1 & 1 & 0 & -2 & 1 \\ 0 & 2 & 1 & 1 & 0 & 0 \end{array} \right) \quad (R_1 \leftarrow R_1 - R_2) \\
 & \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 0 & 3 & -1 \\ 0 & 1 & 1 & 0 & -2 & 1 \\ 0 & 0 & -1 & 1 & 4 & -2 \end{array} \right) \quad (R_3 \leftarrow R_3 - 2R_2) \\
 & \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 0 & 3 & -1 \\ 0 & 1 & 1 & 0 & -2 & 1 \\ 0 & 0 & 1 & -1 & -4 & 2 \end{array} \right) \quad (R_3 \leftarrow -R_3) \\
 & \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -1 & 1 \\ 0 & 1 & 1 & 0 & -2 & 1 \\ 0 & 0 & 1 & -1 & -4 & 2 \end{array} \right) \quad (R_1 \leftarrow R_1 + R_3) \\
 & \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & 1 & 2 & -1 \\ 0 & 0 & 1 & -1 & -4 & 2 \end{array} \right) \quad (R_2 \leftarrow R_2 - R_3)
 \end{aligned}$$

Hence

$$A^{-1} = \begin{pmatrix} -1 & -1 & 1 \\ 1 & 2 & -1 \\ -1 & -4 & 2 \end{pmatrix}.$$

Exercise 4

Apply the method of Gauss–Jordan to

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 3 & 1 \\ 3 & 5 & 0 \end{pmatrix}.$$

Is A invertible?

$$\left(\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 2 & 3 & 1 & 0 & 1 & 0 \\ 3 & 5 & 0 & 0 & 0 & 1 \end{array} \right)$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -1 & 3 & -2 & 1 & 0 \\ 3 & 5 & 0 & 0 & 0 & 1 \end{array} \right) \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -1 & 3 & -2 & 1 & 0 \\ 0 & -1 & 3 & -3 & 0 & 1 \end{array} \right) \quad (R_3 \leftarrow R_3 - 3R_1)$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -1 & 3 & -2 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 1 \end{array} \right) \quad (R_3 \leftarrow R_3 - R_2)$$

A zero row appears in the left block: the method stops and A is **not invertible**.
Indeed, the third row of A is the sum of the first two rows, and

$$\det(A) = 1 \cdot (0 - 5) - 2 \cdot (0 - 3) + (-1) \cdot (10 - 9) = -5 + 6 - 1 = 0.$$

Exercise 5

Consider, for $k \in \mathbb{R}$, the matrix

$$A(k) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & k & 1 \\ 1 & 1 & k^2 \end{pmatrix}.$$

1. For which values of k is $A(k)$ invertible?
2. Compute $A(0)^{-1}$ with the method of Gauss–Jordan.

1. We compute $\det(A(k))$ by Gaussian elimination (row additions only):

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & k & 1 \\ 1 & 1 & k^2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & k-1 & 0 \\ 1 & 1 & k^2 \end{pmatrix} \quad (R_2 \leftarrow R_2 - R_1)$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & k-1 & 0 \\ 0 & 0 & k^2-1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_1)$$

The last matrix is upper triangular, hence

$$\det(A(k)) = 1 \cdot (k-1) \cdot (k^2-1) = (k-1)^2(k+1).$$

Therefore $A(k)$ is invertible if and only if $k \neq 1$ and $k \neq -1$.

2. For $k = 0$ we have $\det(A(0)) = 1$, and:

$$\begin{pmatrix} 1 & 1 & 1 & | & 1 & 0 & 0 \\ 1 & 0 & 1 & | & 0 & 1 & 0 \\ 1 & 1 & 0 & | & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 & | & 1 & 0 & 0 \\ 0 & -1 & 0 & | & -1 & 1 & 0 \\ 1 & 1 & 0 & | & 0 & 0 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - R_1)$$

$$\begin{pmatrix} 1 & 1 & 1 & | & 1 & 0 & 0 \\ 0 & -1 & 0 & | & -1 & 1 & 0 \\ 0 & 0 & -1 & | & -1 & 0 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_1)$$

$$\begin{pmatrix} 1 & 1 & 1 & | & 1 & 0 & 0 \\ 0 & 1 & 0 & | & 1 & -1 & 0 \\ 0 & 0 & -1 & | & -1 & 0 & 1 \end{pmatrix} \quad (R_2 \leftarrow -R_2)$$

$$\begin{pmatrix} 1 & 1 & 1 & | & 1 & 0 & 0 \\ 0 & 1 & 0 & | & 1 & -1 & 0 \\ 0 & 0 & 1 & | & 1 & 0 & -1 \end{pmatrix} \quad (R_3 \leftarrow -R_3)$$

$$\begin{pmatrix} 1 & 0 & 1 & | & 0 & 1 & 0 \\ 0 & 1 & 0 & | & 1 & -1 & 0 \\ 0 & 0 & 1 & | & 1 & 0 & -1 \end{pmatrix} \quad (R_1 \leftarrow R_1 - R_2)$$

$$\begin{pmatrix} 1 & 0 & 0 & | & -1 & 1 & 1 \\ 0 & 1 & 0 & | & 1 & -1 & 0 \\ 0 & 0 & 1 & | & 1 & 0 & -1 \end{pmatrix} \quad (R_1 \leftarrow R_1 - R_3)$$

Hence

$$A(0)^{-1} = \begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix}.$$

Exercise 6

Consider, for $k \in \mathbb{R}$, the matrix $A(k) = \begin{pmatrix} k & 1 \\ 4 & k \end{pmatrix}$. For which values of k is $A(k)$ invertible? For these values, write $A(k)^{-1}$, and compute $A(3)^{-1}$.

We have $\det(A(k)) = k^2 - 4 = (k-2)(k+2)$, so $A(k)$ is invertible if and only if $k \neq 2$ and $k \neq -2$. In this case

$$A(k)^{-1} = \frac{1}{k^2 - 4} \begin{pmatrix} k & -1 \\ -4 & k \end{pmatrix}.$$

For $k = 3$: $\det(A(3)) = 5$ and

$$A(3)^{-1} = \frac{1}{5} \begin{pmatrix} 3 & -1 \\ -4 & 3 \end{pmatrix}.$$

Exercise 7

Compute the inverse of

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix}$$

with the adjugate (cofactor) formula $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$.

Laplace expansion along the first row: $\det(A) = 2 \cdot (1 - 0) - 0 + 1 \cdot (3 - 0) = 5 \neq 0$.
The cofactors $C_{ij} = (-1)^{i+j} \det(A_{ij})$ are:

$$C_{11} = +\det \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} = 1, \quad C_{12} = -\det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = -1, \quad C_{13} = +\det \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} = 3,$$

$$C_{21} = -\det \begin{pmatrix} 0 & 1 \\ 3 & 1 \end{pmatrix} = 3, \quad C_{22} = +\det \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} = 2, \quad C_{23} = -\det \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} = -6,$$

$$C_{31} = +\det \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -1, \quad C_{32} = -\det \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} = 1, \quad C_{33} = +\det \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} = 2.$$

The adjugate matrix is the transpose of the cofactor matrix:

$$\text{adj}(A) = \begin{pmatrix} 1 & -1 & 3 \\ 3 & 2 & -6 \\ -1 & 1 & 2 \end{pmatrix}' = \begin{pmatrix} 1 & 3 & -1 \\ -1 & 2 & 1 \\ 3 & -6 & 2 \end{pmatrix}, \quad A^{-1} = \frac{1}{5} \begin{pmatrix} 1 & 3 & -1 \\ -1 & 2 & 1 \\ 3 & -6 & 2 \end{pmatrix}.$$

Exercise 8

Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$.

1. Compute A^{-1} , B^{-1} and then $(AB)^{-1}$ using the properties of the inverse. Check the result by in-

verting $\mathbf{AB}$ directly.

2. Show that $\mathbf{A}^{-1}\mathbf{B}^{-1} \neq (\mathbf{AB})^{-1}$.
3. Compute $(\mathbf{B}')^{-1}$ and $\det((\mathbf{AB})^{-1})$.

1. $\det(\mathbf{A}) = 1$ and $\det(\mathbf{B}) = 3$, hence

$$\mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{B}^{-1} = \frac{1}{3} \begin{pmatrix} 3 & -2 \\ 0 & 1 \end{pmatrix},$$

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} = \frac{1}{3} \begin{pmatrix} 3 & -2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 5 & -7 \\ -1 & 2 \end{pmatrix}.$$

Check: $\mathbf{AB} = \begin{pmatrix} 2 & 7 \\ 1 & 5 \end{pmatrix}$, $\det(\mathbf{AB}) = 10 - 7 = 3$, and the 2×2 formula gives

$$(\mathbf{AB})^{-1} = \frac{1}{3} \begin{pmatrix} 5 & -7 \\ -1 & 2 \end{pmatrix}.$$

$$2. \quad \mathbf{A}^{-1}\mathbf{B}^{-1} = \frac{1}{3} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 3 & -2 \\ 0 & 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 3 & -3 \\ -3 & 4 \end{pmatrix} \neq (\mathbf{AB})^{-1}.$$

$$3. \quad (\mathbf{B}')^{-1} = (\mathbf{B}^{-1})' = \frac{1}{3} \begin{pmatrix} 3 & 0 \\ -2 & 1 \end{pmatrix} \text{ (check with the } 2 \times 2 \text{ formula applied to } \mathbf{B}' = \begin{pmatrix} 1 & 0 \\ 2 & 3 \end{pmatrix}). \text{ Moreover}$$

$$\det((\mathbf{AB})^{-1}) = \frac{1}{\det(\mathbf{AB})} = \frac{1}{\det(\mathbf{A})\det(\mathbf{B})} = \frac{1}{3}.$$

Exercise 9

1. Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be invertible and $\lambda \in \mathbb{R}$, $\lambda \neq 0$. Show that $\lambda\mathbf{A}$ is invertible and $(\lambda\mathbf{A})^{-1} = \frac{1}{\lambda}\mathbf{A}^{-1}$.
2. Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ satisfy $\mathbf{A}^2 - 3\mathbf{A} + \mathbf{I} = \mathbf{0}$. Show that $\mathbf{A}$ is invertible and $\mathbf{A}^{-1} = 3\mathbf{I} - \mathbf{A}$. Check this on $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$.

1. Using the properties of the product by a scalar:

$$(\lambda \mathbf{A})\left(\frac{1}{\lambda} \mathbf{A}^{-1}\right) = \lambda \cdot \frac{1}{\lambda} \mathbf{A} \mathbf{A}^{-1} = \mathbf{I}, \quad \left(\frac{1}{\lambda} \mathbf{A}^{-1}\right)(\lambda \mathbf{A}) = \frac{1}{\lambda} \cdot \lambda \mathbf{A}^{-1} \mathbf{A} = \mathbf{I}.$$

2. From $\mathbf{A}^2 - 3\mathbf{A} + \mathbf{I} = \mathbf{0}$ we get $3\mathbf{A} - \mathbf{A}^2 = \mathbf{I}$, i.e.,

$$\mathbf{A}(3\mathbf{I} - \mathbf{A}) = \mathbf{I} \quad \text{and} \quad (3\mathbf{I} - \mathbf{A})\mathbf{A} = \mathbf{I}.$$

Hence $\mathbf{A}^{-1} = 3\mathbf{I} - \mathbf{A}$. For $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$:

$$\mathbf{A}^2 = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}, \quad \mathbf{A}^2 - 3\mathbf{A} + \mathbf{I} = \begin{pmatrix} 5-6+1 & 3-3 \\ 3-3 & 2-3+1 \end{pmatrix} = \mathbf{0},$$

and $3\mathbf{I} - \mathbf{A} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$, which is indeed $\mathbf{A}^{-1}$.

Exercise 10

Compute the inverse of the upper triangular matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{pmatrix}$$

with the method of Gauss–Jordan. What kind of matrix is $\mathbf{A}^{-1}$?

The left block is already upper triangular with ones on the diagonal: we only need to eliminate the entries above the pivots, starting from the last column.

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & -3 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right) \quad (R_1 \leftarrow R_1 - 3R_3)$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & -3 \\ 0 & 1 & 0 & 0 & 1 & -4 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right) \quad (R_2 \leftarrow R_2 - 4R_3)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -2 & 5 \\ 0 & 1 & 0 & 0 & 1 & -4 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right) \quad (R_1 \leftarrow R_1 - 2R_2)$$

Hence

$$A^{-1} = \begin{pmatrix} 1 & -2 & 5 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{pmatrix},$$

which is again upper triangular with ones on the diagonal.